

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: multiple CAT and other vectors under different promoters including HIV, SIV, human and rhesus cytomegalovirus immediate early gene, SV40 and the simian retrovirus type 2.
Species Used	Human, HeLa, epithelial carcinoma		

Before the Pulse

Cell Growth Medium	RPMI 1640 with 10 mM dextrose and 0.1mM dithiothreitol (Sigma, GIBCO/ BRL, Flow Labs)	Growth Phase at Harvest	Log phase (We routinely subdivide cells 24 hours prior to electroporation)
		Pre-pulse Incubation	None
Wash Solution	Trypsin harvest, washed twice in Phosphate Buffered Saline		

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	25 °C		
Electroporation Medium*	RPMI 1640 without Fetal Calf Serum , +10mM dextrose, 0.1 mM dithiothreitol	Cuvette Gap	0.4 cm
Cell Density	1.3 x 10 ⁷ (7) viable cells / ml, used 0.3 ml	Voltage	0.300 kV
Volume of Cells	300 µl	Field Strength	0.750 kV/cm
DNA Concentration	500 µg / ml (5 to 10 µg per pulse)		
DNA Resuspension Buffer	Not given	Capacitor	500 µF
Volume of DNA	10 to 20 µl	Resistor	(Pulse Controller) Ω none
After the Pulse		Time Constant	33 to 38 msec
Outgrowth Medium	DMEM culture media, 10% fetal calf serum		

Outgrowth Temperature	37 °C	Relevant Publications and/or Comments Note: exponential values designated in parentheses. All plasmids were prepared by alkaline lysis and banded to equilibrium twice in cesium chloride gradients. Largest DNA molecule electroporated has been the rhesus cytomegalovirus genome (approx. 225 kB). In general, the higher the efficiency of transfection, the higher degree of cell killing. For some cell types, > 90% of the input cells are killed, although > 90% of the survivors express very high levels of input DNA. We have found that ≥ 20 µg of DNA per electroporation resulted in greatly reduced efficiency of transfection. Similarly, 2.5 µg of DNA resulted in reduced efficiency.
Length of Incubation	48 hours	
Selection Method or Assay Used	Transient (CAT, b-gal, immunohisto-chemistry).	
Electroporation Efficiency	50 to100%	
Per Cent Survival	20 to 75%	

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094